

PROJECT PROPOSAL
COMPUTER ORGANIZATION AND ASSEMBLY LANGUAGE
(COAL)
BCS 3J



Title:

SNAKE GAME

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Project's Overview :

This project focuses on developing a classic Snake Game using Assembly Language. The Snake Game is a retro arcade-style game where the player controls a snake that moves around the screen, eating food to grow longer while avoiding collisions with its own tail or the screen boundaries. The objective is to achieve the highest possible score by eating as much food as possible. This project emphasizes the application of low-level programming skills, memory management, and efficient hardware interaction.

Goals and Objectives :

- **Enhance understanding of Assembly Language :** Gain in depth knowledge of low-level programming concepts, including memory management, and performance optimization.
- **Core Gameplay Mechanics :** Implement snake movement in all four directions (Up, Down, Left, and Right). Ensure smooth and responsive gameplay mechanics.
- **User Input Handling :** Manage keyboard inputs to control the snake's direction using W,A,S,D keys.
- **Scoring System :** Implement a real-time scoring system that increments as the snake eats food and displays the score on the screen.
- **Game Over Conditions:** Detect collisions with the snake's body or screen boundaries to trigger a game-over state.
- **Difficulty Level :** The game will have three levels of difficulty: easy, medium or hard depending on the speed of the snake which can be slow, medium or hard.

Project Components :

1. W,A,S,D as control keys
2. Snake movement
3. Real time Scoring and Timing
4. Difficulty Level
5. Game Over Conditions
6. User Interface
7. Retry

Technologies :

Programming Language : Assembly Language

Assembler : MASM

Expected Outcome :

1. Functional Gameplay:

- A fully functional Snake Game with real-time snake movement, food generation, and collision detection.

2. User Interaction:

- Smooth and responsive keyboard inputs to control the snake's direction.

3. Graphical Output:

- A colorful and visually appealing game screen with real-time score updates.

4. COAL Fundamentals:

- Application of loops, jumps, stacks, and variable declarations to manage game mechanics and state.

5. Performance Optimization:

- An efficient and low-resource implementation showcasing the benefits of low-level programming.

Conclusion :

The Snake Game project serves as a practical application of Computer Organization and Assembly Language (COAL) concepts. By implementing gameplay mechanics such as snake movement, collision detection, and scoring, the project highlights key programming principles like loops, jumps, and memory management.

This project provides hands-on experience in low-level game development, demonstrating how assembly language can optimize performance and hardware interaction. The successful implementation of the Snake Game showcases a deeper understanding of programming efficiency and the fundamental mechanics of real-time applications, making it a valuable step toward mastering assembly language and exploring game development at a hardware-interaction level.